AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2015

General Mathematics Paper II

Time: 2 hours 20 minutes Marks: 45

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are TEN questions. Answer ALL Questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue or correcting fluid.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

(ATTEMPT EITHER PART a OR PART b OF Q.1.)

Q.1. (Total 5 Marks)

a.

i. If $p+q=7$ and $p-q=1$, then find the value of pq. (3 Marks)

ii. Without using calculator, simplify $5\sqrt{24}-\sqrt{6}$. (2 Marks)

b. If $\frac{1}{x^2} = 2 - \sqrt{3}$, then find the value of $x^2 + \frac{1}{x^2}$. (5 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.2.)

Q.2. (Total 5 Marks)

a. A polynomial $P(x)$ is defined as $P(x) = x^3 + kx^2 - 3x - 6$.

i. When $P(x)$ is divided by $(x-2)$, the remainder is 4. Find the value of *k*. (2 Marks)

ii. If $(x-2)$ and $(ax^2 - 3)$ are factors of $P(x)$, find the value of *a*. (3 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.2.)

b. Factorize $x^3 + y^3 - (x+y)^2 - x - y$. (5 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.3.)

Q.3. (Total 4 Marks)

a. If $4 - 12x + ax^2$ is a perfect square, find the value of *a* using **division method**.

b. The highest common factor and the least common multiple of TWO quadratic expressions are $(5x+4)$ and $(3x+4)(2x-5)(5x+4)$ respectively. Find these two quadratic expressions.

Q.4. (Total 5 Marks)

Solve $|4x - 1| = |3x - 2|$. Also verify your answer.

(ATTEMPT EITHER PART a OR PART b OF Q.5.)

Q.5. (Total 5 Marks)

a. Use **quadratic formula** to find the values of x for $16x^2 + 28x = 4$.

b. The sum of squares of two consecutive positive odd numbers is 34. Find the smaller of the two numbers.

(ATTEMPT EITHER PART a OR PART b OF Q.6.)

Q.6. (Total 5 Marks)

a. The matrices A and B are defined as $A = \begin{bmatrix} 1 & 6 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. Verify that commutative law under multiplication does not hold for matrices A and B. (5 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.6.)

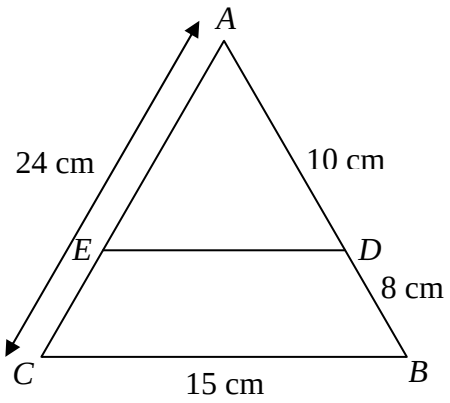
b. If $A^{-1} = \begin{bmatrix} 1 & -1 \\ 2 & 2.5 \end{bmatrix}$ and $|A| = 2$, find (5 Marks)

i. matrix *A*. (4 Marks)

ii. transpose of matrix *A*. (1 Mark)

Q.7. (Total 5 Marks)

In the given diagram triangle *ABC* is similar to triangle *ADE*.



NOT TO SCALE

If $\overline{AD} = 10\,\text{cm}$, $\overline{DB} = 8\,\text{cm}$, $\overline{BC} = 15\,\text{cm}$ and $\overline{AC} = 24\,\text{cm}$, find the length of

i. $\overline{AE}$ (3 Marks)

ii. $\overline{ED}$ (2 Marks)

Q.8. (Total 4 Marks)

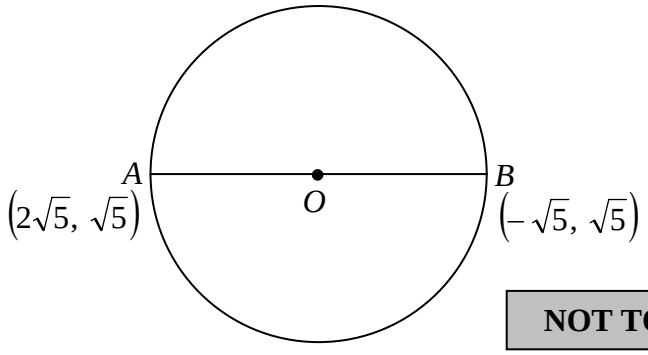
Draw a parallelogram *ABCD* such that $\overline{AB} = 8\text{cm}$, $\overline{BC} = 5\text{cm}$ and $\angle ABC = 140°$.

Q.9. (Total 4 Marks)

Find the area of a square that has a diagonal of 10 cm.

Q.10. (Total 3 Marks)

A circle having centre *O* is shown below. Find the coordinates of point *O*.



NOT TO SCALE

END OF PAPER